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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 19	
TITLE: JDBC SELECT Operation	

Problem Statement

Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives

To establish database connectivity and retrieve records using JDBC SELECT statements.

1. Create a Java application to display student records from database using SELECT query.

Programme

Welcome StudentSelectDemo.java

StudentSelectDemo.java > StudentSelectDemo > main(String[])

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class StudentSelectDemo {
    Run | Debug
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/rikishab";
        String user = "root";
        String password = "YourNewPassword123!";
        String query = "SELECT roll_no, name, course, marks FROM students";

        try {
            Class.forName(className: "com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(query);
            System.out.println(x: "--- Student Records ---");
            while (rs.next()) {
                int rollNo = rs.getInt(columnLabel: "roll_no");
                String name = rs.getString(columnLabel: "name");
                String course = rs.getString(columnLabel: "course");
                double marks = rs.getDouble(columnLabel: "marks");

                System.out.println("Roll No: " + rollNo + ", Name: " + name +
                                   ", Course: " + course + ", Marks: " + marks);
            }

            rs.close();
            stmt.close();
            conn.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>  c:; cd 'c:\Users\RIKISHA\PROGRAMMING\Java\src'; & 'C:
Temp\cp_eoalbjrpbr5egr0wencuomc3p.argfile' 'StudentSelectDemo'
--- Student Records ---
Roll No: 101, Name: Aarav Sharma, Course: Computer Science, Marks: 88.5
Roll No: 102, Name: Rikisha Bhattacharya, Course: Information Technology, Marks: 94.0
Roll No: 103, Name: Rohan Verma, Course: Data Science, Marks: 82.0
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. Create a Java application that connects to a database and displays product details using a SELECT query. Show product ID, product name, quantity, and price.

Programme

Welcome | StudentSelectDemo.java | ProductSelectDemo.java

ProductSelectDemo.java > ProductSelectDemo > main(String[])

```java
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class ProductSelectDemo {
    Run | Debug
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/rikishab";
        String user = "root";
        String password = "YourNewPassword123!";

        String query = "SELECT product_id, product_name, quantity, price FROM products";

        try {
            Class.forName(className: "com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(url, user, password);
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery(query);

            System.out.println(x: "--- Product Inventory ---");
            while (rs.next()) {
                int id = rs.getInt(columnLabel: "product_id");
                String name = rs.getString(columnLabel: "product_name");
                int quantity = rs.getInt(columnLabel: "quantity");
                double price = rs.getDouble(columnLabel: "price");

                System.out.println("ID: " + id + " | Name: " + name +
                                   " | Quantity: " + quantity + " | Price: $" + price);
            }

            rs.close();
            stmt.close();
            conn.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>  c:; cd 'c:\Users\RIKISHA\PROGRAMMING\Java\s
Temp\cp_eoalbjrpbr5egr0wencuomc3p.argfile' 'ProductSelectDemo'
--- Product Inventory ---
ID: 101 | Name: Wireless Mouse | Quantity: 150 | Price: $25.5
ID: 102 | Name: Mechanical Keyboard | Quantity: 85 | Price: $75.99
ID: 103 | Name: 27-inch Monitor | Quantity: 40 | Price: $299.0
ID: 104 | Name: USB-C Docking Station | Quantity: 120 | Price: $45.75
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

Conclusion:

Thus, the JDBC SELECT operation was successfully implemented to establish database connectivity and retrieve student and product records using SQL SELECT queries. This assignment helped in understanding how Java interacts with a database and displays the retrieved data using JDBC.